The Tubulin-GUE Conjecture

A Falsifiable Prediction Connecting Microtubule Excitonic Dynamics to Random Matrix Theory

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Abstract

This paper presents a novel, falsifiable prediction: that the temporal intervals between excitonic state transitions in microtubule tryptophan networks follow the nearest-neighbor spacing distribution (NNSD) of the Gaussian Unitary Ensemble (GUE) from random matrix theory. An exhaustive literature review confirms that no published work (2003-2026) has applied Wigner surmise or any RMT spacing distribution to biological event-timing intervals at the molecular scale. All prior RMT applications in biology have analyzed eigenvalue spectra of correlation matrices, a fundamentally different application. The prediction rests on three empirical pillars: room-temperature ultraviolet superradiance from tryptophan mega-networks in microtubules (Babcock et al., 2024), anesthetic disruption of excitonic transport (Kalra et al., 2023), and the mature mathematical framework of quantum chaos crossovers (BGS conjecture, Brody interpolation). Specific numerical thresholds are provided: the mean consecutive spacing ratio should yield r = 0.603 +/- 0.02 (GUE) under physiological conditions and shift toward r = 0.386 (Poisson) under anesthetic administration. The prediction requires a minimum of 500 independent switching events for initial discrimination and 2,000 for precise crossover characterization. Detailed experimental protocols using single-molecule FRET and tryptophan fluorescence lifetime spectroscopy are specified. This conjecture, if validated, would constitute the first empirical evidence of quantum chaotic dynamics in a biological molecular system and establish a direct link between the Montgomery-Odlyzko law governing Riemann zeta zero spacings and biological information processing.

Keywords: *random matrix theory, Gaussian Unitary Ensemble, microtubules, tubulin, tryptophan superradiance, quantum biology, Wigner surmise, level spacing statistics, falsifiable prediction, Montgomery-Odlyzko law*

1. Introduction

The Gaussian Unitary Ensemble (GUE) of random matrix theory occupies a singular position in mathematical physics. Its nearest-neighbor spacing distribution governs the statistical behavior of energy levels in quantum chaotic systems, the spacings between non-trivial zeros of the Riemann zeta function, and the eigenvalue correlations of Hermitian matrices drawn from unitarily invariant probability measures [1, 2]. The universality of GUE statistics across these domains suggests that it encodes a fundamental principle of correlated information-bearing systems. This paper proposes extending this universality to biological molecular dynamics by predicting that excitonic state transition intervals in microtubule tryptophan networks follow GUE nearest-neighbor spacing statistics.

The prediction is motivated by three recent experimental advances. First, Babcock et al. (2024) demonstrated ultraviolet superradiance from networks of over 10^5 tryptophan $1L_a$ transition dipoles in microtubule architectures at room temperature, confirming collective quantum optical behavior in the cytoskeleton [3]. Second, Kalra et al. (2023) showed that anesthetics isoflurane and etomidate reduce exciton diffusion in microtubules by approximately 20%, with an exciton diffusion length of 6.6 nm exceeding Forster theory predictions by a factor of five [4]. Third, Khan et al. (2024) demonstrated that microtubule-stabilizing drugs delay anesthetic-induced unconsciousness in rats (Cohen's d = 1.9), providing in vivo evidence linking microtubule dynamics to consciousness [5].

The central question is whether the temporal correlations among excitonic transition events in microtubules exhibit the hallmark signature of quantum chaos: level repulsion following GUE statistics rather than the uncorrelated Poisson statistics expected from classical stochastic processes. An exhaustive review of the literature confirms that this question has never been asked. All published applications of random matrix theory in biology (approximately 20 papers, 2003-2026) analyze eigenvalue spectra of correlation or covariance matrices. None have applied Wigner surmise or any RMT spacing distribution to temporal intervals between biological events at the molecular scale. This paper fills that gap with a precisely specified, falsifiable prediction.

2. Mathematical Framework

2.1 The GUE Wigner Surmise

The Wigner surmise for the GUE (symmetry parameter beta = 2) gives the nearest-neighbor spacing distribution for 2 x 2 Hermitian matrices drawn from the unitarily invariant Gaussian measure. Let s denote the spacing between consecutive events normalized to unit mean. The distribution is:

$$p_{GUE}(s) = (32/\text{pi}^2)\, s^2 \exp(-4s^2/\text{pi}) \quad (1)$$

with normalization constants $a_2 = 32/\text{pi}^2 = 3.2423$ and $b_2 = 4/\text{pi} = 1.2732$. The quadratic onset $p(s) \sim s^2$ for small s reflects level repulsion: transitions are suppressed from occurring too close together. This contrasts sharply with the Poisson distribution p(s) = exp(-s), which has maximal probability at s = 0 and describes uncorrelated events. The spacing variance for GUE is 0.180, compared to 0.273 for GOE and 1.000 for Poisson [6, 7].

The exact GUE spacing distribution for large-N matrices involves the Fredholm determinant of the sine kernel K(x,y) = sin(pi(x-y))/(pi(x-y)). The gap probability is $E_2(0;s) = \det(I - K|_{L2(0,s)})$, expressible via a Painleve V transcendent [8]. For practical application to biological data, the Wigner surmise (Eq. 1) provides an approximation accurate to within 5% in the bulk and 10% in the tails, sufficient for the sample sizes achievable in single-molecule experiments.

2.2 The Ratio of Consecutive Spacings

The most robust discriminator between ensembles is the ratio of consecutive spacings $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$, introduced by Oganesyan and Huse (2007) and formalized by Atas et al. (2013) [9]. This statistic requires no spectral unfolding, eliminating the most significant source of systematic error in RMT analysis. The exact Wigner surmise for ratios is:

$$P_W^{\text{beta}}(r) = C_{\text{beta}}\, (r + r^2)^{\text{beta}} / (1 + r + r^2)^{1+3\text{beta}/2} \quad (2)$$

yielding mean values: $r_{\text{Poisson}} = 0.386$, $r_{\text{GOE}} = 0.536$, $r_{\text{GUE}} = 0.603$. These values are cleanly separated and constitute the primary numerical prediction of this paper [9].

2.3 Connection to the Montgomery-Odlyzko Law

The theoretical anchor connecting GUE statistics to fundamental mathematics is Montgomery's pair correlation conjecture (1973): the normalized spacings between non-trivial zeros of the Riemann zeta function zeta(s) on the critical line Re(s) = 1/2 follow the GUE pair correlation function $R_2(u) = 1 - (\sin(\pi u)/(\pi u))^2$ [2]. Odlyzko's numerical verification extended to billions of zeros at heights up to 10^{23}, with agreement to within 0.002 [10]. Rodgers (2024) proved that the GUE Conjecture for zeta zeros is logically equivalent, conditional on the Riemann Hypothesis, to a statement about prime distribution [11]. If tubulin excitonic transitions follow GUE statistics, this would establish a direct empirical link between biological information processing and the same mathematical structure governing the distribution of prime numbers.

3. The Biological Substrate

3.1 Tryptophan Superradiance Networks

Each tubulin dimer contains 8 tryptophan residues whose $1L_a$ transition dipoles participate in collective quantum optical behavior. Babcock et al. (2024) demonstrated that networks of over 10^5 tryptophan dipoles in microtubule bundles produce superradiant states emitting in hundreds of femtoseconds, while subradiant states persist for tens of seconds, spanning many orders of magnitude in lifetime [3]. The effect survives thermal disorder at five times room temperature and was selected as a Science Editors' Choice. The computed tubulin dimer dipole moment is approximately 1,740 Debye.

The physical mechanism underlying the prediction is as follows. In a quantum chaotic system, the eigenstates of the Hamiltonian are delocalized and exhibit correlations described by random matrix theory. The Bohigas-Giannoni-Schmit (BGS) conjecture (1984) states that quantum systems whose classical limit is chaotic follow RMT statistics, while the Berry-Tabor conjecture (1977) predicts Poisson statistics for integrable systems [12, 13]. The tryptophan excitonic Hamiltonian in a microtubule lattice, coupling over 10^5 dipoles through a disordered but structured lattice, is expected to exhibit quantum chaotic dynamics if quantum coherence is maintained over biologically relevant timescales. The resulting energy level spacings, and by extension the temporal intervals between excitonic state transitions, would then follow GUE statistics due to the broken time-reversal symmetry inherent in dissipative biological environments.

3.2 Timescale Hierarchy

Tubulin dynamics span multiple timescales, and the prediction must specify which transitions are being tested. Molecular dynamics simulations from the Voth group show local intradimer rotation angle interconversions occurring on timescales of tens of nanoseconds [14]. The full straight-to-kinked conformational transition involves barriers that place the switching time above 10 microseconds [15]. GTP hydrolysis in the microtubule lattice proceeds at approximately 0.2 per second [16]. The Bandyopadhyay group reports electromagnetic resonance frequencies spanning MHz to GHz ranges, with conformational-type oscillations at 0.8-8 MHz and electronic/dipolar oscillations at 30-40 GHz [17, 18].

For the present conjecture, the relevant timescale is the tryptophan excitonic transition dynamics: superradiant lifetimes of hundreds of femtoseconds to nanoseconds, and subradiant lifetimes extending to seconds. The prediction specifically targets the intervals between successive excitonic state transitions within the tryptophan network of individual tubulin dimers or small microtubule segments, measurable via time-resolved fluorescence spectroscopy.

4. The Tubulin-GUE Conjecture: Formal Statement

Conjecture (Papiri, 2026). Let $\{t_1, t_2, ..., t_N\}$ be the ordered sequence of excitonic state transition times in a microtubule tryptophan network under physiological conditions (37 degrees C, aqueous buffer, pH 7.4). Define the spacing sequence $s_n = t_{n+1} - t_n$ and normalize to unit mean. Then the nearest-neighbor spacing distribution of $\{s_n\}$ converges to the GUE Wigner surmise (Eq. 1) as N approaches infinity, with the mean consecutive spacing ratio satisfying *r* = 0.603 +/- 0.02.

Corollary (Anesthetic Decoherence Shift). Under administration of volatile anesthetics (isoflurane, halothane) at clinically relevant concentrations (1-2 MAC), the mean consecutive spacing ratio shifts from *r* = 0.603 (GUE) toward *r* = 0.386 (Poisson), the Brody parameter q decreases from approximately 2 toward 0, and the spacing variance increases from approximately 0.18 toward 1.0. This shift reflects the disruption of quantum coherence in the tryptophan excitonic network by anesthetic binding to hydrophobic pockets near aromatic residues, as experimentally demonstrated by Kalra et al. (2023) [4].

Table 1. *Quantitative predictions for spacing statistics under physiological and anesthetic conditions.*

Statistic	GUE Prediction (Physiological)	Poisson Prediction (Anesthetic)	GOE (Intermediate)
Mean spacing ratio *r*	0.603 +/- 0.02	0.386 +/- 0.02	0.536 +/- 0.02
Brody parameter q	~2.0	~0.0	~1.0
Spacing variance	0.180	1.000	0.273
Number variance $\mathrm{Sigma}^2(10)$	0.56	10.0	1.09
Level repulsion exponent	beta = 2 (quadratic)	beta = 0 (none)	beta = 1 (linear)
Spectral rigidity $\mathrm{Delta}_3(L)$	Logarithmic	Linear (L/15)	Logarithmic

5. Novelty: The Confirmed Literature Gap

An exhaustive search of the published literature (2003-2026) confirms that no paper has applied Wigner surmise or any GUE/GOE spacing distribution to biological conformational switching intervals, ion channel dwell times, or single-molecule event timing data. This represents a genuine first-mover opportunity. The existing RMT applications in biology fall into three categories, none of which address the present question.

First, network eigenvalue analysis: Kikkawa (2018) applied GOE/Poisson NNSD analysis to eigenvalues of gene interaction network adjacency matrices in cancer cells, confirming GOE statistics for dense networks via Kolmogorov-Smirnov tests [19]. Seba (2003) analyzed EEG cross-correlation matrix eigenvalue spectra [20]. Montgomery (2024) reviewed Marchenko-Pastur and Wigner semicircle laws as null models for neural connectivity matrices [21]. Second, protein dynamics covariance: Potestio, Caccioli, and Vivo (2009) fitted the Brody distribution (beta approximately 0.8) to eigenvalue spacings of protein dynamics covariance matrices from molecular dynamics simulations [22]. Third, the sole partial exception: Martinis et al. (2003) applied Brody distribution fitting to heartbeat RR intervals, finding only local agreement with RMT predictions and characterizing the dynamics as mixed type under Berry-Robnik theory [23]. This preprint generated no follow-up.

The conceptual innovation of the present conjecture lies in treating a sequence of molecular conformational switching intervals as analogous to nuclear energy level spacings and testing for level repulsion. Single-molecule dwell times are universally analyzed using exponential distributions, hidden Markov models, gamma distributions, or dispersed kinetics frameworks. The question of whether these intervals exhibit the correlated spacing statistics characteristic of quantum chaos has never been posed at the molecular scale.

6. Experimental Protocol

6.1 Primary Method: Tryptophan Fluorescence Lifetime Spectroscopy

The primary experimental approach exploits the tryptophan superradiance phenomenon characterized by Babcock et al. (2024) [3]. Time-resolved fluorescence measurements of tryptophan emission from microtubule preparations, using time-correlated single-photon counting (TCSPC) with approximately 25 ps instrument response, can resolve individual excitonic state transitions. The protocol involves preparing taxol-stabilized microtubules from purified tubulin, exciting tryptophan residues at 280 nm, and recording emission at 340 nm with single-photon sensitivity. Individual photon arrival times encode the excitonic transition dynamics of the tryptophan network.

The Stern-Volmer quenching analysis framework developed by Kalra et al. (2023) provides a complementary readout [4]. By measuring the exciton diffusion coefficient D and diffusion length L_D = 6.6 nm under controlled conditions, one obtains a statistical ensemble of excitonic transport events. The key measurement is the distribution of inter-event intervals in the fluorescence intensity record, extracted via change-point detection algorithms.

6.2 Complementary Method: Single-Molecule FRET on Tubulin

Single-molecule FRET has not yet been performed on individual tubulin dimers but is technically feasible. A 2023 international blind study across 19 laboratories demonstrated FRET efficiency uncertainty of 0.06 or less, corresponding to interdye distance precision within 2 angstroms [24]. Confocal smFRET achieves microsecond-to-millisecond temporal resolution, and photon-by-photon H2MM analysis can detect conformational transitions down to approximately 150 microseconds [25]. Site-specific labeling of tubulin at two positions flanking the intradimer interface would enable direct observation of conformational switching events.

6.3 Anesthetic Perturbation Protocol

The anesthetic decoherence prediction (Corollary) requires repeating the measurements under controlled anesthetic conditions. Isoflurane at 1-2 MAC equivalent concentration (0.3-0.6 mM in solution) should be applied to the microtubule preparation. Kalra et al. (2023) demonstrated that this concentration reduces exciton diffusion by approximately 20% and shortens tryptophan fluorescence lifetimes proportionally [4]. A structurally related non-anesthetic anticonvulsant serves as a negative control, as it does not produce the same effect on excitonic transport.

Table 2. *Minimum experimental requirements.*

Parameter	Requirement	Justification
Switching events (N)	>= 500 (initial); >= 2,000 (crossover)	KS test at alpha = 0.05 for GUE vs. Poisson
Temporal resolution	<= 25 ps (TCSPC)	Resolve superradiant fs-ns dynamics
Temperature	37 C +/- 0.5 C	Physiological; effect survives 5x RT
Anesthetic concentration	1-2 MAC (0.3-0.6 mM isoflurane)	Clinically relevant; matches Kalra et al.

MT stabilization	Taxol or Epothilone B	Prevent depolymerization artifacts

7. Falsification Criteria

A prediction without falsification criteria is not science. The following criteria specify the exact conditions under which the Tubulin-GUE Conjecture is disproved, supported, or inconclusive.

7.1 The Conjecture Is Disproved If

All four conditions are simultaneously met: (i) the mean consecutive spacing ratio satisfies $r < 0.45$; (ii) the Brody parameter from maximum likelihood estimation satisfies $q < 0.3$; (iii) the Kolmogorov-Smirnov test against the Poisson distribution yields $p > 0.10$; and (iv) the spacing variance exceeds 0.7. This outcome would demonstrate that tubulin excitonic transitions are uncorrelated classical events with no quantum coherence signature, consistent with the Tegmark decoherence objection [26].

7.2 The Conjecture Is Supported If

All four conditions are simultaneously met: (i) $r > 0.55$; (ii) $q > 0.7$; (iii) the number variance $\text{Sigma}^2(L)$ grows logarithmically rather than linearly for L = 1 to 20; and (iv) the spacing variance is below 0.25. If additionally $r > 0.58$ and the log-log slope of P(s) for $s < 0.3$ yields an exponent within 0.3 of beta = 2, this constitutes strong evidence for GUE specifically rather than GOE.

7.3 The Conjecture Is Inconclusive If

Results fall in the intermediate range: $0.45 < r < 0.55$ with $0.3 < q < 0.7$. This would indicate partial quantum coherence, potentially consistent with semi-Poisson statistics ($P(s) = 4s \exp(-2s)$, r approximately 0.50) associated with Anderson metal-insulator transitions. Such an outcome would be scientifically valuable in its own right, indicating a quantum-classical boundary regime in biological molecular dynamics [27].

7.4 Systematic Error Controls

Several systematic effects must be controlled. Incorrect spectral unfolding can generate spurious level repulsion, creating false positives. This risk is mitigated by using the consecutive spacing ratio r, which requires no unfolding. Missing levels or symmetry mixing push statistics toward Poisson and cannot create false positives for quantum coherence. Non-stationarity (drift in transition rates) must be tested via sliding-window analysis. Berry-Robnik mixing of regular and chaotic phase-space regions would produce intermediate statistics that must be deconvolved [28].

8. Statistical Analysis Protocol

The recommended analysis pipeline proceeds in eight steps. Step 1: collect a minimum of 500 switching intervals from the fluorescence record. Step 2: compute the mean consecutive spacing ratio r as the primary discriminator, requiring no unfolding. Step 3: perform careful spectral unfolding using a smoothed staircase function of the cumulative distribution. Step 4: construct the NNSD histogram with approximately 30-50 bins and overlay the GUE, GOE, and Poisson theoretical curves. Step 5: fit the Brody distribution $P(s) = (q+1)b\ s^q \exp(-b\ s^{q+1})$ via maximum likelihood estimation to extract q. Step 6: perform Kolmogorov-Smirnov tests against each of the three hypotheses. Step 7: compute the number variance $\text{Sigma}^2(L)$ for L ranging from 0.5 to 20. Step 8: assess the log-log behavior of P(s) for $s < 0.3$ to extract the level repulsion exponent beta [6, 9].

For the anesthetic crossover experiment, the Brody parameter q should be tracked as a function of anesthetic concentration, producing a decoherence curve. The transition from GUE (q approximately 2) to Poisson (q = 0) as a function of anesthetic concentration constitutes the most informative test of the conjecture. The theoretical framework for this crossover is provided by the model Hamiltonian H(lambda) = (H_0 + lambda V) / sqrt(1 + lambda^2), which interpolates between Poisson (lambda = 0) and GUE (lambda approaching infinity), with transition parameter proportional to lambda^{2}d, where d is a coupling parameter [29].

9. Theoretical Implications

9.1 For Quantum Biology

Validation of the Tubulin-GUE Conjecture would constitute the first empirical demonstration of quantum chaotic dynamics at the molecular scale in a biological system. While quantum effects in biology have been demonstrated in photosynthetic light harvesting, avian magnetoreception, and enzymatic tunneling, none of these involve quantum chaos as characterized by RMT spacing statistics. The conjecture extends the quantum biology program from coherent transport phenomena to the deeper regime of universal spectral fluctuations.

9.2 For the Orch-OR Theory of Consciousness

The Penrose-Hameroff Orchestrated Objective Reduction theory posits that consciousness arises from quantum processes in microtubules [30]. A demonstration that microtubule excitonic dynamics follow GUE statistics would provide independent evidence that quantum coherence is maintained in the microtubule system at biologically relevant temperatures. The anesthetic decoherence prediction (Corollary) directly tests the Orch-OR mechanism: if anesthetics shift tubulin spacing statistics from GUE toward Poisson, this links the loss of consciousness to the loss of quantum coherence, precisely as Orch-OR predicts.

9.3 For the Riemann Hypothesis

The Montgomery-Odlyzko law establishes that Riemann zeta zero spacings follow GUE statistics. If tubulin excitonic transitions independently follow the same statistics, this would be the first physical system below the nuclear scale to exhibit GUE universality in event-timing data. While this does not constitute progress toward proving the Riemann Hypothesis, it would provide a new biological testbed for studying GUE universality and potentially suggest that the mathematical structure underlying the distribution of primes has a wider physical instantiation than previously recognized [2, 10, 11].

9.4 For Neuromorphic Architecture Design

The N-BIO (Neuromorphic Biological Integration) architecture proposes implementing sub-neuronal computation via cellular automata emulating microtubule dynamics [31]. If tubulin switching follows GUE statistics, this provides a precise mathematical constraint on the statistical properties that the N-BIO Digital Biopolymer Layer must reproduce. Specifically, the transition rules of the Protofilament Cellular Automata should be designed such that their temporal output statistics match the GUE Wigner surmise, not Poisson or GOE distributions. This constitutes a falsifiable design criterion for neuromorphic hardware claiming biological fidelity at the sub-neuronal scale.

10. Conclusion

The Tubulin-GUE Conjecture proposes a specific, falsifiable prediction that has never been tested: that excitonic state transition intervals in microtubule tryptophan networks follow GUE nearest-neighbor spacing statistics. The prediction is grounded in experimentally demonstrated quantum optical behavior of microtubules (superradiance, anomalous exciton diffusion, anesthetic modulation), supported by the mature mathematical framework of quantum chaos, and occupies a confirmed gap in the scientific literature. The numerical predictions are precise: $r = 0.603$ under physiological conditions, shifting toward 0.386 under anesthesia. The experimental methodology exists at the required temporal resolution. The falsification criteria are sharp.

If validated, this conjecture would simultaneously advance quantum biology, the Orch-OR theory of consciousness, our understanding of GUE universality, and the design of biologically faithful neuromorphic architectures. If disproved, it would establish a clear negative result constraining the quantum coherence hypothesis for microtubules. Either outcome represents a contribution to science. The experiment should be performed.

References

[1] Mehta, M. L. (2004). Random Matrices, 3rd ed. Academic Press.

[2] Montgomery, H. L. (1973). The pair correlation of zeros of the zeta function. Proc. Symp. Pure Math. 24, 181-193.

[3] Babcock, N. S., Montes-Cabrera, G., Oberhofer, K. E., et al. (2024). Ultraviolet Superradiance from Mega-Networks of Tryptophan in Biological Architectures. J. Phys. Chem. B 128(17), 4035-4046. DOI: 10.1021/acs.jpcb.3c07936.

[4] Kalra, A. P., Benny, A., Travis, S. M., et al. (2023). Electronic Energy Migration in Microtubules. ACS Central Science 9(3), 352-361. DOI: 10.1021/acscentsci.2c01114.

[5] Khan, S. M., et al. (2024). Microtubule-stabilizer Epothilone B delays isoflurane-induced loss of righting reflex in rats. eNeuro 11(8), ENEURO.0291-24.2024. DOI: 10.1523/ENEURO.0291-24.2024.

[6] Haake, F. (2010). Quantum Signatures of Chaos, 3rd ed. Springer.

[7] Shir, R., Martinez-Azcona, P., and Chenu, A. (2025). Surmise for random matrices' level spacing distributions beyond nearest-neighbors. arXiv: 2504.20134.

[8] Jimbo, M., Miwa, T., Mori, Y., and Sato, M. (1980). Density matrix of an impenetrable Bose gas and the fifth Painleve transcendent. Physica D 1(1), 80-158.

[9] Atas, Y. Y., Bogomolny, E., Giraud, O., and Roux, G. (2013). Distribution of the Ratio of Consecutive Level Spacings in Random Matrix Ensembles. Phys. Rev. Lett. 110, 084101.

[10] Odlyzko, A. M. (2001). The 10^{22}-nd zero of the Riemann zeta function. Contemp. Math. 290, 139-144.

[11] Rodgers, B. (2024). Arithmetic Consequences of the GUE Conjecture for Zeta Zeros. Michigan Math. J. 74(5), 899-976. DOI: 10.1307/mmj/20216139.

[12] Bohigas, O., Giannoni, M. J., and Schmit, C. (1984). Characterization of chaotic quantum spectra and universality of level fluctuation laws. Phys. Rev. Lett. 52, 1-4.

[13] Berry, M. V. and Tabor, M. (1977). Level clustering in the regular spectrum. Proc. R. Soc. London A 356, 375-394.

[14] Gebremichael, Y., Chu, J.-W., and Voth, G. A. (2008). Intrinsic bending and structural rearrangement of tubulin dimer. Biophys. J. 95, 2487-2499.

[15] Igaev, M. and Gruebmuller, H. (2018). Microtubule assembly governed by tubulin allosteric gain in flexibility and lattice induced fit. eLife 7, e34353.

[16] Roostalu, J., et al. (2020). The speed of GTP hydrolysis determines GTP cap size and controls microtubule stability. eLife 9, e51992.

[17] Sahu, S., Ghosh, S., Hirata, K., Fujita, D., and Bandyopadhyay, A. (2013). Multi-level memory switching in single brain microtubules. Appl. Phys. Lett. 102, 123701.

[18] Saxena, K., et al. (2020). Fractal, Scale Free Electromagnetic Resonance of a Single Brain Extracted Microtubule Nanowire, a Single Tubulin Protein and a Single Neuron. Fractal Fract. 4(2), 11.

[19] Kikkawa, A. (2018). Random Matrix Analysis for Gene Interaction Networks in Cancer Cells. Sci. Rep. 8, 10607.

[20] Seba, P. (2003). Random Matrix Analysis of Human EEG Data. Phys. Rev. Lett. 91, 198104.

[21] Montgomery, D. (2024). Random matrix theory approaches to neural connectivity. WNSC 2(1), 1-22.

[22] Potestio, R., Caccioli, F., and Vivo, P. (2009). Random Matrix Approach to Collective Behavior and Bulk Universality in Protein Dynamics. Phys. Rev. Lett. 103, 268101.

[23] Martinis, M., Mikuta-Martinis, V., Stehlik, A., and Knezevic, A. (2003). Spectral Statistics of RR Intervals in ECG. arXiv: physics/0301011.

[24] Agam, G., et al. (2023). Reliability and accuracy of single-molecule FRET studies. Nature Methods. DOI: 10.1038/s41592-023-01807-0.

[25] Lerner, E., Cordes, T., Ingargiola, A., et al. (2018). Toward dynamic structural biology: Two decades of single-molecule Forster resonance energy transfer. Science 359, eaan1133.

[26] Tegmark, M. (2000). Importance of quantum decoherence in brain processes. Phys. Rev. E 61, 4194-4206.

[27] Bogomolny, E. B., Gerland, U., and Schmit, C. (2001). Short-range plasma model for intermediate spectral statistics. Eur. Phys. J. B 19, 121-127.

[28] Berry, M. V. and Robnik, M. (1984). Semiclassical level spacings when regular and chaotic orbits coexist. J. Phys. A: Math. Gen. 17, 2413.

[29] Chavda, N. D., et al. (2014). Poisson to GOE transition in the distribution of the ratio of consecutive level spacings. Phys. Lett. A 378, 3012-3017.

[30] Hameroff, S. and Penrose, R. (2014). Consciousness in the universe: A review of the 'Orch OR' theory. Phys. Life Rev. 11(1), 39-78.

[31] Papiri Melendez, L. G. (2026). Neuromorphic Biological Integration Architecture (N-BIO): Multi-Scale Bio-Inspired Computing with Sub-Neuronal Processing. Unpublished manuscript.